

AI Mentor Architecture Blueprint

Goal: Build a production-minded AI voice mentor starting with Deep Learning and expanding later.

Finalized Tech Stack

Frontend: Next.js, TypeScript, Tailwind CSS.
Backend: Python, FastAPI, Uvicorn, Pydantic.
LLM: Qwen 3 (GGUF).
Inference: llama.cpp (development), vLLM (future).
Embeddings: BAAI bge-base-en-v1.5.
Vector DB: Qdrant.
Speech-to-Text: faster-whisper.
Text-to-Speech: Piper.
Database: PostgreSQL.
PDF Processing: PyMuPDF.
Containerization: Docker (Qdrant initially; all services later).

High-Level Request Flow

User Voice → faster-whisper → FastAPI → Embedding Model (question) → Qdrant
retrieves relevant chunks → Qwen 3 via llama.cpp → Piper TTS → User hears answer.

Development Roadmap

1. Build chat UI.
2. Integrate FastAPI backend.
3. Run Qwen 3 locally with llama.cpp.
4. Build PDF ingestion pipeline.
5. Generate embeddings and store in Qdrant.
6. Implement RAG retrieval.
7. Add speech-to-text.
8. Add text-to-speech.
9. Add authentication and PostgreSQL.
10. Dockerize services and prepare for production.

Suggested Folder Structure

```
frontend/  
  app/  
  components/  
  services/
```

```
backend/  
  main.py  
  api/  
  services/  
  rag/  
  voice/  
  models/  
  db/  
  utils/  
  data/
```

Future Evolution

Replace llama.cpp with vLLM for GPU serving. Keep Qwen model family, Qdrant, FastAPI APIs, and frontend unchanged.